



Department of Education

**STEM Education**  
**National Schools of Excellence**  
**Biology Teacher's Resource Book**  
**Grade 12**  
**Term 1**

**DOMAIN**      **Phylum**

**SCIENCE**      **Biology**

**Technology**  
**Engineering**  
**Mathematics**

Issued free to schools by the Department of Education

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### **Acknowledgements**

This STEM Biology Syllabus for **Grade 12** constitutes ten (10) Thematic Areas (Thema) and is intended for students enrolling for STEM Education. In this booklet only **six** Thema are presented.

Ultima Thulé Ltd was engaged by the Department of Education to develop the STEM Curriculum. The content was generated by writers from universities, secondary schools, new university graduates with knowledge in the subject matter and industry subject experts. Quality assurance groups and the Science Subject Advisory Committee have also contributed to the development of this syllabus.

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## Table of Contents

|                                                                           |    |
|---------------------------------------------------------------------------|----|
| SECRETARY’S MESSAGE .....                                                 | 4  |
| INTRODUCTION .....                                                        | 4  |
| RATIONALE .....                                                           | 7  |
| NEW NOMENCLATURE.....                                                     | 8  |
| STEM Domains.....                                                         | 9  |
| STEM BIOLOGY CONTENT OVERVIEW – TERM 2 .....                              | 9  |
| Table 1.1 Sequencing & Benchmarking STEM Chemistry Content Standards..... | 10 |
| Table 1.2 Mapping Content Progressions by Modules.....                    | 14 |
| THEMA: BIOTECHNOLOGY .....                                                | 15 |
| MODULE 12.1: BIOINFORMATICS.....                                          | 17 |
| Topic 1: Transcription and Translation .....                              | 18 |
| Topic 2: DNA Sequence Data Analysis .....                                 | 20 |
| Topic 3: Bioinformatics.....                                              | 21 |
| Topic 4: DNA Extraction and Sequencing .....                              | 22 |
| THEMA: HEALTH AND MEDICINE.....                                           | 25 |
| MODULE 12.1: HEALTH AND MEDICAL BIOLOGY .....                             | 27 |
| Topic 1: Homeostasis.....                                                 | 28 |
| Topic 2: Disease – Disruption of Homeostasis .....                        | 29 |
| Topic 3: Systemic Response to Disease or Injury .....                     | 30 |
| Topic 4: Surgery: Dissection of Animals.....                              | 32 |

## SECRETARY'S MESSAGE

This Biology Syllabus is to be used exclusively by teachers of Biology to teach Grade 11 Students enrolling for the STEM Education through the Schools of Excellence throughout Papua New Guinea. The STEM Biology Bridging Syllabus ensured that Grade 11 students enrolling for STEM Education were grounded in the basic concepts espoused in Biology before undertaking the advanced concepts introduced in this syllabus.

The syllabus conforms to the visions of the National Education Plan and the School of Excellence Policy. Furthermore, it is undergirded by a policy and legislative matrix that were issued by the National Government over the last decades, such as the Vision 2050 and Development Strategic Plan 2030. The Preamble of the National Constitution, especially the First Directive of the *National Goals and Directive Principles* makes a clarion call for the Integral Human Development, which the Department took heed of to ensure, through this new syllabus, that *all forms of beneficial creativity, including sciences and cultures, [are] to be actively encouraged*. A principle object of the *National Education Act 1983 (amended 1995)* reiterates this virtue of Integral Human Development (Section 4).

In accordance with the widespread demand for a home-grown curriculum to replace the existing one with the view to position the country as a bastion of innovation in the new century the Secretary of Education has embarked on introducing the STEM Curriculum to the Schools of Excellence. It is hoped that students coming through the STEM Education will not only gain productive employment in the knowledge-based economy but become creators, innovators and inventors of knowledge and ultimately commercialize their creativity through development of their intellectual property rights.

In addition, there are four core elements espoused in this syllabus, insofar as student development is concerned: Students are expected to gain knowledge and skills and apply them to solve every day problems whilst appreciating Biology as a fundamental field of science.

This STEM Biology Syllabus for Grade 11 constitutes ten (10) Thematic Area (Thema) and is intended for students enrolling for STEM Education. In this booklet only three Thema are presented. By studying these Thema students are expected to gain the foundational knowledge needed in Biology.

The success of this new Curriculum is highly dependent on the Teachers. They play a pivotal role by being innovative and keeping abreast of new information on scientific research and innovations. The challenge for teachers of Biology is to engage students to learn realistic context for increased and better understanding and appreciate the relevance of Biology to human lives, including good health and environment.

In order to ensure Teachers discharge their responsibilities adequately this syllabus is accompanied by a Teacher's Guide and Teacher's Resource Book which provides sufficient teaching and assessment materials to help Teachers.

I commend and approve this syllabus as the official curriculum for Biology to be used for the Grade 11 STEM Education in the Schools of Excellence throughout Papua New Guinea.

DR UKE KOMBRA, PHD  
Secretary for Education

## INTRODUCTION

Biology is the study of living organisms and how they interact at multiple levels to sustain life. The basic building block of life is considered to be the cell. The hereditary units are the genetic sequences, and these are encased within the cells. This genetic information ensures the hereditary traits necessary for survival of the species is passed on from parents to progenies. Biology is divided into different fields of study. For instance, botany is the study of plants whilst zoology involves the study of animals. Micro-organisms are studied in microbiology. Recent fields that emerged in the last few decades include molecular biology, genomics, and so on.

Since Biology is an experimental science, laboratory work becomes an essential part of the syllabus. Practical work can be used to enable students to investigate the properties and reactions of substances and to provide opportunities for students to learn and test the principles and concepts of Biology. Since Biology is a highly specialized subject students undertaking Biology are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills.

The modules in this Syllabus are sequenced to guide teachers of Biology through the process of imparting basic Biology knowledge and skills to students.

The **Grade 12** Biology Syllabus entails **six (6)** Thematic Areas (Thema) and these are listed below. Each Thema has a single Module and **four (4)** topics. Over time different Modules will be introduced.

| THEMA               | MODULE                           | TOPICS                                                                                                                                                                                                    |
|---------------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BIOTECHNOLOGY       | 12.1:<br>Bioinformatics          | 1. Transcription and Translation<br>2. How to Read Gene Sequence<br>3. Gene Design<br>4. DNA Extraction and Sequencing                                                                                    |
| HEALTH AND MEDICINE | 12.1: Health and Medical Biology | 1. Homeostasis<br>2. Disease – Disruption of Homeostasis<br>3. Systemic Response to Disease or Injury<br>4. Biology of Hemostasis<br>5. Basics of Disease Diagnosis<br>6. Surgery – Dissection of Animals |
| ENERGY              | 12.1: Bioenergy                  | 1. Energy Conversion by Photosynthetic Organisms<br>2. Production of Fuel Alcohol from Cellulosic Biomass<br>3. Biomass and Biofuels<br>4. Biogas Production                                              |
| BATTERY TECHNOLOGY  | 12.1: Bio-Batteries              | 1. Introduction to Biobatteries<br>2. Paper batteries<br>3. Sugar Batteries<br>4. Bioelectrochemical Reactor<br>5. The Moses Experiment – Burning Bush                                                    |

|                         |                                           |                                                                                                                                                                                                                                                                                                              |
|-------------------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ARTIFICIAL INTELLIGENCE | 12.1: Neuroscience                        | 1. General Design of the Nervous System<br>2. Sensory Receptors and Neuronal Circuits for Processing Information<br>3. Sense – The Eye and Vision<br>4. Cerebral Cortex, Intellectual Functions of Brain, Learning, and Memory<br>5. Behavioral and Motivational Mechanisms of the Brain – The limbic System |
| ROBOTICS                | 12.1: Biological Inspirations to Robotics | 1. Motor and Integrative Neurophysiology<br>2. Sensory Systems<br>3. Biologically Inspired Robots<br>4. Cyborg                                                                                                                                                                                               |

Since Biology is a highly specialized subject students undertaking Biology are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills.

The modules in this Syllabus are sequenced to guide teachers of Biology through the process of imparting basic Biology knowledge and skills to students.

# RATIONALE

There is a confluence of events that necessitated the need to introduce STEM Curriculum to the Schools of Excellence:

1. The School of Excellence Policy that was launched by Prime Minister Honourable James Marape, MP, in June 2020, calls for the establishment of the Schools of Excellence and the introduction of STEM Curriculum;
2. The Government has decided to replace the Observation Based Education to Standards Based Education in 2019 and tasked the Department of Education to introduce the STEM Curriculum as part of its reform agenda;
3. There is public outcry, both from public and private sectors, over the quality of graduates coming through the National Education System; and
4. The changing landscape in the technology domain globally and the pressing need to produce the next generation of workforce that is agile and can adopt to the changing circumstances as well as become enterprising individuals through their creation of intellectual properties.

The combination of these factors has led the Department of Education to adopt the STEM Curriculum and to introduce this to the Schools of Excellence as a necessary disruptive intervention in an effort to develop the next generation of national intellectual capital and to drive the growth and prosperity of the country through creation of intellectual properties.

In developing the STEM Curriculum, it was realized that the Grades 9 and 10 Science Syllabi does not provide a firm foundation to launch students into STEM Curriculum at the Schools of Excellence. Consequently, a STEM Bridging Program was necessary. The STEM Bridging Syllabus is a bridge across the divide that Grade 11 Students enrolling for STEM Education must cross.

In the Biology Bridging Syllabus students were introduced to the basic concepts and knowledge of Biology in order for them to develop an appreciation of the discipline in a broader context. In this syllabus students are introduced to advanced concepts in Biology as they are applied to the ten Thema.

The introduction of the STEM Curriculum, especially the Bridging Program, will cause consequential changes to the Upper Classes of STEM Education and the Lower Secondary Syllabi. These syllabi will be introduced in the course of time.

## NEW NOMENCLATURE

As part of the overall Curriculum Development new nomenclature were introduced. These are explained below:

| Term   | Explanation                                                                                                                                           |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Domain | Each STEM area is defined as <i>domain</i> (e.g. Science Domain, Technology Domain, etc.)                                                             |
| Phylum | What would normally be regarded as “subject” (e.g. Biology, Physics, Chemistry, etc.) is now regarded as <i>phylum</i> ; the plural is <i>phyla</i> . |
| Thema  | Thematic areas of studies                                                                                                                             |
| Module | A specific concept within the Thema which is being studied                                                                                            |
| Topic  | A specific aspect of the Module that is being studied                                                                                                 |



## **STEM Domains**

At this juncture of our national development there is no other task before us more important than deploying a robust STEM Curriculum that develops the next generation of highly educated students to enter the work force. By building on the best of current practices and standards the STEM Curriculum aims to take us, beyond the constraints of the existing structures of schooling, towards a shared vision of excellence nationally and marketing the nation globally in terms of Science, Technology, Engineering and Mathematics. In this STEM Curriculum, there are ten (10) Thema (Thematic Areas) introduced.

### **Science**

It is the study of the natural world, including the laws of the nature associated with physics, chemistry and biology and the treatment or application of facts, principles, concepts or conventions associated with these disciplines. Science is both a body of knowledge that has been accumulated over time and a process-scientific inquiry-that generates new knowledge. Knowing from science informs the engineering design process. In this STEM Curriculum the Science Domain has three Phyla – Biology, Chemistry and Physics.

### **Technology**

Comprises the entire system of people and organizations, knowledge, processes, and devices that go into creating and operating technological artifacts, as well as the artifacts themselves. Throughout history humans have created technology to satisfy their wants and needs. Much of modern technology is the product of science and engineering and technological tools used in both fields. There are two Phyla – Information Technology and Computer Sciences – introduced in this STEM Curriculum for the Technology Domain.

### **Engineering**

Concerns both the knowledge about the design and creation of products and a process for solving problems. The design process must result in a final product that must operate or function under various constraints. One constraint in engineering design is the laws of nature. Engineers must learn to harness the natural forces at work around them and through their design process develop products that must operate in harmony with nature. Other constraints include things such as time, money, available materials, ergonomics, environmental regulations and manufacturability. Engineering utilizes concepts in science and mathematics as well as technological tools. There are three (3) Phyla introduced in the Engineering Domain – Civil, Mechanical and Electrical.

### **Mathematics**

It is the study of patterns and relationships among quantities, numbers, and shapes. Specific branches of mathematics include arithmetic, geometry, algebra, trigonometry and calculus. Mathematics is used in science and in engineering. In this STEM Curriculum, Mathematics is considered as a Service Domain; specific topics tailored towards specific Thema are also considered relevant to undergird the relevance and applicability of Mathematics Domain to other fields of knowledge.

## **STEM BIOLOGY CONTENT OVERVIEW – TERM 1**

**Table 1.1 Sequencing & Benchmarking STEM Chemistry Content Standards**

| Setting STEM Chemistry Content Standards for Progressive Learning & Assessment |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                            |                                       |
|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| Thema                                                                          | Broad Content Standard                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                            |                                       |
| Biotechnology                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                            |                                       |
| <b>Module 12.1</b><br>Bioinformatics                                           | <b>12.1a:</b> Evaluate and articulate the transcription and translation of DNA into proteins.<br><b>12.1b:</b> Appreciate and describe how a gene sequence can be read.<br><b>12.1c:</b> Discuss the process of designing a gene.<br><b>12.1d:</b> Evaluate and describe the process of DNA extraction, amplification by PCR and DNA <b>12.1e:</b> sequence analysis.                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                            |                                       |
| Topics                                                                         | Performance Indicators<br>Students are expected to:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Module Benchmark                                                                                                                                                                                                           | Assessment Tasks                      |
| 1. Transcription and Translation                                               | <b>12.1.1a:</b> Appreciate and describe the DNA replication process.<br><b>12.1.1b:</b> Identify and describe the process of transcription of DNA into mRNA and translation into amino acids, and final assembly of proteins.<br><b>12.1.1c:</b> Evaluate and analyze the DNA replication process.<br><b>12.1.1d:</b> Describe the transcription and translation of DNA into amino acids and assembly of proteins<br><b>12.1.1e:</b> Appreciate the fundamentals of DNA replication, transcription and translation into amino acids and assembly forming proteins.<br><b>12.1.1f:</b> Appreciate this fundamental concept and their applications in biotechnology. | At the end of this module, students should master the ability, knowledge, and skills to: <ul style="list-style-type: none"> <li>• Read Gene Sequence</li> <li>• Design Gene</li> <li>• Extract and Sequence DNA</li> </ul> | Research Project<br>+<br>Written Test |
| 2. How to Read Gene Sequence                                                   | <b>12.1.2a:</b> Identify and describe the DNA sequencing methods<br><b>12.1.2b:</b> Describe the sequence assembly and                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                            |                                       |

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
|                                  | <p>alignment from available databases</p> <p><b>12.1.2c:</b> Identify and predict genome sequences utilizing a variety of methods</p> <p><b>12.1.2d:</b> Read genome sequences and analyze them</p> <p><b>12.1.2e:</b> Identify the varying sequencing and analyzing methods</p> <p><b>12.1.2f:</b> Able to predict a gene sequence using different methods of genome annotation</p> <p><b>12.1.2g:</b> Able to read and interpret a gene sequence</p> <p><b>12.1.2h:</b> Appreciate the different methods utilized to sequence and analyze a gene</p> <p><b>12.1.2i:</b> Develop this scientific interest into genome editing or genetic engineering</p> |  |  |
| 3. Gene Design                   | <p><b>12.1.3a:</b> Describe and analyze the standard methods of how DNA can be synthesized</p> <p><b>12.1.3b:</b> Analyze and appreciate how gene designs can be constructed for various purposes in genome research.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |  |
| 4. DNA Extraction and Sequencing | <p><b>12.1.4a:</b> Appreciate and describe the DNA extraction processes and why the different reagents were utilized.</p> <p><b>12.1.4b:</b> Appreciate and describe PCR in DNA amplification and the use of gel electrophoresis for analysis.</p> <p><b>12.1.4c:</b> Appreciate the process involved in DNA sequencing and analysis.</p>                                                                                                                                                                                                                                                                                                                 |  |  |

|  |                                                                                                                                                                                                         |  |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
|  | <b>12.1.4d:</b> Consolidate the basic processes in DNA analysis from extraction, amplification by PCR and analysis by gel electrophoresis and finally understanding the basis of the sequence of genes. |  |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|

| Thema                                             | Broad Content Standard                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                    |                                                                    |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Health and Medicine                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                    |                                                                    |
| <b>Module 12.1</b><br>Health and Medicine Biology | <b>12.1a:</b> Identify and describe the parameters and mechanisms that regulate homeostasis within the body.<br><b>12.1b:</b> Evaluate and describe the disease processes within the human body that disrupts homeostasis.<br><b>12.1c:</b> Identify and describe the pathway and level of health care in Papua New Guinea<br><b>12.1d:</b> Appreciate and describe personalized medicine as the future trend of medical and health care model.<br><b>12.1e:</b> Appreciate the underlying concepts and future technologies in health and medical care. |                                                                                                                                                                                                                                                                                                                                    |                                                                    |
| Topics                                            | Performance Indicators<br>Students are expected to:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Module Benchmark                                                                                                                                                                                                                                                                                                                   | Assessment Tasks                                                   |
| 1. Homeostasis                                    | <b>12.1.1a:</b> Evaluate and analyze the importance of maintaining homeostasis within the body.<br><b>12.1.1b:</b> Describe how these vital parameters contribute to regulating the bodies homeostasis                                                                                                                                                                                                                                                                                                                                                  | Student's ability to master the knowledge and skills of: <ul style="list-style-type: none"> <li>• Homeostasis</li> <li>• Disruption of Homeostasis cause disease</li> <li>• Systemic Response to Injury</li> <li>• Biology of Hemostasis</li> <li>• Basis of Disease Diagnosis</li> <li>• Surgery via Animal Dissection</li> </ul> | Quiz<br>+<br>Written Test<br>+<br>Assignment<br>+<br>Lab Practical |
| 2. Disease – Disruption of Homeostasis            | <b>12.1.2a:</b> Outline the clinical significance of what may go wrong in the body that disrupts maintenance of homeostasis resulting in fatal outcomes.<br><b>12.1.2b:</b> Describe how diabetes as example of failed homeostatic control.<br><b>12.1.2c:</b> Analyze and describe how the different variables involved in homeostatic control mechanisms can be measured to determine homeostasis                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                    |                                                                    |

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| 3. Systemic Response to Disease or Injury | <p><b>12.1.3a:</b> Describe how the body responds to disease processes or injuries.</p> <p><b>12.1.3b:</b> Appreciate the immune and hematological system in responding to disease and injury state</p>                                                                                                                                                                                                                                                               |  |  |
| 4. Biology of Hemostasis                  | <p><b>12.1.4a:</b> Explain what Vascular Constriction is</p> <p><b>12.1.4b:</b> Define Platelet and explain its functions</p> <p><b>12.1.4c:</b> Explain what Coagulation is</p> <p><b>12.1.4c:</b> Describe what Fibrinolysis is</p>                                                                                                                                                                                                                                 |  |  |
| 5. Basics of Disease Diagnosis            | <p><b>12.1.5a:</b> Display clinical skills in a practical ( or hypothetical) medical scenario</p> <p><b>12.1.5b:</b> Conduct a physical examination in a practical ( or hypothetical) medical scenario</p> <p><b>12.1.5c:</b> Display and work with principles of Patient Care in a practical ( or hypothetical) medial scenario</p> <p><b>12.1.5d:</b> Explain the Era of Genomics and how it has contributed to the globalization of medicine</p>                   |  |  |
| 6. Surgery – Dissection of Animals        | <p><b>12.1.6a:</b> Dissect an animal and label the structure and function of it correctly.</p> <p><b>12.1.6b:</b> Distinguish the major systems and their functional relationships in correlation to human body</p> <p><b>12.1.6c:</b> Appreciate the similarities and differences in structure and function of the dissected rat and a human body</p> <p><b>12.1.6d:</b> Appreciate the use of surgical tools in dissection and hands on technique in stitching.</p> |  |  |

**Table 1.2 Mapping Content Progressions by Modules**

| Thema/Module                     | Broad Content Standard                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 12.1 | 12.1 |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|
| Biotechnology/ Module 12.1       | <b>12.1a:</b> Evaluate and articulate the transcription and translation of DNA into proteins.<br><b>12.1b:</b> Appreciate and describe how a gene sequence can be read.<br><b>12.1c:</b> Discuss the process of designing a gene.<br><b>12.1d:</b> Evaluate and describe the process of DNA extraction, amplification by PCR and DNA<br><b>12.1e:</b> sequence analysis.                                                                                                                                                                                | ✓    |      |
| Health and Medicine/ Module 12.1 | <b>12.1a:</b> Identify and describe the parameters and mechanisms that regulate homeostasis within the body.<br><b>12.1b:</b> Evaluate and describe the disease processes within the human body that disrupts homeostasis.<br><b>12.1c:</b> Identify and describe the pathway and level of health care in Papua New Guinea<br><b>12.1d:</b> Appreciate and describe personalized medicine as the future trend of medical and health care model.<br><b>12.1e:</b> Appreciate the underlying concepts and future technologies in health and medical care. |      | ✓    |

# **THEMA: BIOTECHNOLOGY**

## Introduction

Biotechnology combines biology with technology to engineer, modify or upgrade the parts of biological system for industrial and human welfare. It combines various technologies derived from biology, mathematics, physics, chemistry, and engineering to modify the biological systems. Its application ranges from agriculture to industry, medicine, nutrition, environmental conservation, cell biology, and so on. Biotechnology is now one of the fastest growing fields of sciences that is solving global issues and at the same time improving standard of living.

In the previous module, introduction to the fundamental concepts of biotechnology and its applications in the medical or pharmaceutical, agricultural, and industrial fields were briefly covered. It has contributed largely to the production of supplementary health care products, drugs, vaccines (e.g., Polio and SARS-CoV2-2019 vaccines), and many more. The most important application is in genetic engineering or recombinant DNA technology utilized to produce genetically modified crops and organisms. These applications in agricultural biotechnology aids to increase crop yields, prevent damage from pests and diseases, and reduce environmental impacts by farming. Further applications in industries include biorefineries built around the world to produce biofuels refined from biomass, that is in order to mitigate greenhouse gases and address Global Warming and Climate Change.

Without the multidisciplinary domain of **Bioinformatics**, none of the above advancements in biotechnology would be feasible. To some degree this will be discussed in this module.



## MODULE 12.1: BIOINFORMATICS

4-5 Weeks

Data management and proper analysis is key to biotechnology. The human genome project and other organism-specific sequencing projects have generated an unprecedented amount of biological data. Bioinformatics, an ever-evolving science, is in charge of meeting the enormous demand for data analysis and interpretation. The application of computation and analysis tools to capture and interpret biological data is referred to as **bioinformatics**. Computer science, mathematics, physics, and biology are all combined in this field. In modern biology and medicine, bioinformatics is required for data management.

Many areas of biology have embraced bioinformatics. Bioinformatics techniques like image and signal processing enable useful results to be extracted from large amounts of raw data in experimental molecular biology. It assists in the sequencing and annotation of genomes and their reported mutations in the area of genetics. It helps to organize and query biological data through text mining of biological literature and the establishment of biological and gene ontologies. It also helps with gene and protein expression and regulation analysis.

### Linking Modules

Linked to

- Chemistry/ Module 12.1: Protein Extraction and Sequencing (*Linked with chemistry phyla for Protein Sequencing. The same plant material (containing cyclotides) utilized can be used for DNA extraction and sequencing as well.*)

### Broad Content Standards

**12.1a:** Evaluate and articulate the transcription and translation of DNA into proteins.

**12.1b:** Appreciate and describe how a gene sequence can be read.

**12.1c:** Discuss the process of designing a gene.

**12.1d:** Evaluate and describe the process of DNA extraction, amplification by PCR and DNA

**12.1e:** sequence analysis.

### Content Benchmarks

At the end of this module, students should master the ability, knowledge, and skills to:

- Read Gene Sequence
- Design Gene
- Extract and Sequence DNA

## Topic 1: Transcription and Translation

### Performance Indicators

Students are expected to;

**12.1.1a:** Appreciate and describe the DNA replication process.

**12.1.1b:** Identify and describe the process of transcription of DNA into mRNA and translation into amino acids, and final assembly of proteins.

**12.1.1c:** Evaluate and analyze the DNA replication process.

**12.1.1d:** Describe the transcription and translation of DNA into amino acids and assembly of proteins

**12.1.1e:** Appreciate the fundamentals of DNA replication, transcription and translation into amino acids and assembly forming proteins.

**12.1.1f:** Appreciate this fundamental concept and their applications in biotechnology.

### Classroom Exercise

#### Activities

- Describe the Information Storage in Biology. This includes;
  - Basic DNA Structure and the four complementary base nucleotides
  - How DNA Replication occurs
  - What happens in a PCR
- Describe the Information Flow in Biology
  - Explain what the Genetic Code is?
  - Describe the basic structure of Proteins and how they are formed?
  - Complete the table by labelling the different codons and their corresponding protein encoded.

| 1 <sup>st</sup> Position | 2 <sup>nd</sup> Position |   |   |   | 3 <sup>rd</sup> Position |
|--------------------------|--------------------------|---|---|---|--------------------------|
|                          | U                        | C | A | G |                          |
| U                        |                          |   |   |   | U<br>C<br>A<br>G         |
| C                        |                          |   |   |   | U<br>C<br>A<br>G         |
| A                        |                          |   |   |   | U<br>C<br>A<br>G         |
| G                        |                          |   |   |   | U<br>C<br>A<br>G         |
|                          | Amino Acids              |   |   |   |                          |

Ala: Alanine

Gln: Glutamine

Leu: Leucine

Ser: Serine

Asn: Asparagine

Glu: Glutamic acid

Lys: Lysine

Thr: Threonine

Arg: Arginine

Gly: Glycine

Met: Methionine

Trp: Tryptophan

Asp: Aspartic Acid

His: Histidine

Phe: Phenylalanine

Tyr: Tyrosine

Cys: Cysteine

Ile: Isoleucine

Pro: Proline

Val: Valine

- Describe the different levels of protein structure and their fundamental differences
- How is the flow of Information in Biology controlled?
  - Explain the transcriptional control
  - Explain the translational Control
  - Describe how RNA Regulation occurs in the information flow

## Topic 2: DNA Sequence Data Analysis

### Performance Indicators

Students are expected to:

- 12.1.2a:** Identify and describe the DNA sequencing methods
- 12.1.2b:** Describe the sequence assembly and alignment from available databases
- 12.1.2c:** Identify and predict genome sequences utilizing a variety of methods
- 12.1.2d:** Read genome sequences and analyze them
- 12.1.2e:** Identify the varying sequencing and analyzing methods
- 12.1.2f:** Able to predict a gene sequence using different methods of genome annotation
- 12.1.2g:** Able to read and interpret a gene sequence
- 12.1.2h:** Appreciate the different methods utilized to sequence and analyze a gene
- 12.1.2i:** Develop this scientific interest into genome editing or genetic engineering

### Classroom Exercise

#### Activity 1:

- What is DNA Sequencing?
- History of Sequencing
  - Who developed the idea of Protein and DNA Sequencing and won two noble prizes?
  - Why was the initial method of sequencing?
- Describe what whole Genome Sequencing is and sequences are assembled
- Describe what Shotgun Sequencing is?
- What are the main Sequence Assembly Problems?
- Explain how Sequence Alignment is done?
- Describe basically what Next-Generation Sequencing is and what it does?
- Give examples of the different modern sequencing technologies

## Topic 3: Bioinformatics

### Performance Indicators

Students are expected to:

**12.1.3a:** Describe and analyze the standard methods of how DNA can be synthesized

**12.1.3b:** Analyze and appreciate how gene designs can be constructed for various purposes in genome research.

### Classroom Exercise

#### Activity:

- Discuss why bioinformatics is important in Biotechnology?
- Describe the process of finding genes
  - Identify and describe the sense and antisense strands
  - What is an ORF and state its function?
  - Explain how computer programs read the DNA sequence?
  - Describe what introns and exons are?
  - Explain what alternative splicing is?
- Bioinformatics: Finding Functions
  - After going through the example of Finding Functions in Bioinformatics, use blast to isolate, sequence and match genes. It would be fun to do that in class with students in groups.
  - Added also could be to isolate variations in DNA or amino acid sequences by using CLUSTAL

## Topic 4: DNA Extraction and Sequencing

### Performance Indicators

Students are expected to:

- 12.1.4a:** Appreciate and describe the DNA extraction processes and why the different reagents were utilized.
- 12.1.4b:** Appreciate and describe PCR in DNA amplification and the use of gel electrophoresis for analysis.
- 12.1.4c:** Appreciate the process involved in DNA sequencing and analysis.
- 12.1.4d:** Consolidate the basic processes in DNA analysis from extraction, amplification by PCR and analysis by gel electrophoresis and finally understanding the basis of the sequence of genes.

### Laboratory Experiment

**This is an interphylum activity relating physics and chemistry in terms of DNA or genome sequencing and analysis.**

- Sample Preparation
- How Biologist Read a Gene with DNA Sequencing
- Carry Out the Experiment by Isolating and Extracting a Cyclotide DNA molecule if equipped with the necessary materials. Otherwise, video sessions of DNA extraction from various youtube or scientific educational sites would be recommended.
- PCR amplification and analysis of gel could be outsourced to University of Technology or medical school (UPNG campus) where a PCR thermocycler and reagents are available.
- DNA sequencing of the selected gel could be sent for sequencing out of the country, PNG as we do not have a genome sequencing facility.
- Analysis and interpretation of the DNA sequence is key to understanding all this processes. Finally, consultation for more information would be recommended for interested students.
- Note that there are several DNA extractions protocols available for varying kits.

# SAMPLE ASSESSMENT

## Sample Assessment Methods

- Group Assignment
- Presentation
- Projects
- Written or Practical Quiz
- Written or Practical Test
- Practical Assignment
- Case Studies

## Sample Assessment Tasks

| Module                          | Content Standard                                                                                                                                                                                                                                                                                                                                                         | Topic                                                                                                                                                  | Assessment Tasks               |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>12.1:<br/>Bioinformatics</b> | <b>12.1a:</b> Evaluate and articulate the transcription and translation of DNA into proteins.<br><b>12.1b:</b> Appreciate and describe how a gene sequence can be read.<br><b>12.1c:</b> Discuss the process of designing a gene.<br><b>12.1d:</b> Evaluate and describe the process of DNA extraction, amplification by PCR and DNA<br><b>12.1e:</b> sequence analysis. | <b>1.</b> Transcription and Translation<br><b>2.</b> DNA Sequence Data Analysis<br><b>3.</b> Bioinformatics<br><b>4.</b> DNA Extraction and Sequencing | Project +<br>Written Quiz/Test |

## Sample Assessment Marking Guide

| Topic                                                                                           | Assessment Task    | Assessment Criteria                                                            | Total Marks |
|-------------------------------------------------------------------------------------------------|--------------------|--------------------------------------------------------------------------------|-------------|
| 3.<br>Proteomics                                                                                | Written Quiz       | What is Bioinformatics?                                                        | 2.5         |
|                                                                                                 |                    | How is proteomics and genomics related?                                        | 2.5         |
|                                                                                                 |                    | What is the difference between a transcription and a Translation?              | 4           |
|                                                                                                 |                    | List at least three applications of biophysics and briefly explain them.       | 6           |
|                                                                                                 |                    | List the main ingredients used in Sanger sequencing.                           | 2.5         |
|                                                                                                 |                    | Briefly describe the Edman Degradation process.                                | 2.5         |
| Subtotal                                                                                        |                    |                                                                                | 20          |
| Note that this is similar to the Experiment given in GR 12 STEM Chemistry; THEMA: BIOTECHNOLOGY |                    |                                                                                |             |
| Module 12.1<br>(All three Topics)                                                               | Project/Experiment | Purify Protein Sequencing                                                      | 20          |
|                                                                                                 |                    | Determine the amino acid sequence of a protein                                 | 20          |
|                                                                                                 |                    | Check if the protein sequence exist on a database of protein sequences online. | 10          |
| Subtotal                                                                                        |                    |                                                                                | 50          |

|                                   |                                                                             |                                          |            |
|-----------------------------------|-----------------------------------------------------------------------------|------------------------------------------|------------|
| Module 12.1<br>(All three Topics) | Written Test<br><i>(Multiple Choices, Short Answers, Calculations, etc)</i> | Correctly answers all long questions     | 10         |
|                                   |                                                                             | Correctly answers all multiple choice    | 15         |
|                                   |                                                                             | Correctly answers short answer questions | 5          |
|                                   |                                                                             | <b>Subtotal</b>                          | <b>30</b>  |
| <b>Module TOTAL</b>               |                                                                             |                                          | <b>100</b> |



# **THEMA: HEALTH AND MEDICINE**

## **Introduction**

Medicine is the art, science and practice of caring for a patient and managing the diagnosis, prognosis, prevention, treatment or palliative care of their injury or diseases. Medicine encompasses health care practices evolved to maintain and restore health by the prevention and treatment of illness. The practice of medicine has significantly changed with the advent of molecular biology with its enormous implications in biological sciences (sequencing of human genome), sophisticated new imaging techniques and advances in bioinformatics and information technology. That has sparked an explosion of scientific information that fundamentally influenced and changed the way in which patient care and management has been practiced.

The knowledge gleaned from the science of medicine has already improved and undoubtedly will further improve our understanding of complex disease processes and provide new approaches to disease treatment and prevention. Though the practice of medicine requires a scientific basis, the combination of medical knowledge, intuition, experience, and judgment defines the art of medicine.

## MODULE 12.1: HEALTH AND MEDICAL BIOLOGY

*4-5 Weeks*

Health and medical biology are fields related to the discipline of biology that is concerned with medicine, healthcare and diagnostics. An individual functions normally when homeostatic mechanisms are fully functional to maintain a constant internal environment. However, disease processes cause disruption to these homeostatic mechanisms resulting in illness. Individuals who are ill often seek medical attention and health care facilities for management and treatment of their illness. Hence, a dynamic and convenient medical and health care system would be effective in managing patient welfare. Therefore, newly introduced technologies such as personalized medicine are on the verge of changing the way healthcare has been delivered.

### Linking Modules

- STEM Bridging Biology

### Broad Content Standard

**12.1a:** Identify and describe the parameters and mechanisms that regulate homeostasis within the body.

**12.1b:** Evaluate and describe the disease processes within the human body that disrupts homeostasis.

**12.1c:** Identify and describe the pathway and level of health care in Papua New Guinea

**12.1d:** Appreciate and describe personalized medicine as the future trend of medical and health care model.

**12.1e:** Appreciate the underlying concepts and future technologies in health and medical care.

### Content Benchmark

Student's ability to master the knowledge and skills of:

- Homeostasis
- Disruption of Homeostasis cause disease
- Systemic Response to Injury
- Surgery via Animal Dissection

## Topic 1: Homeostasis

### Performance Indicators

Students are expected to:

**12.1.1a:** Describe homeostasis and the feedback mechanisms.

**12.1.1b:** Describe and outline the clinical significance of the different homeostatic parameters

**12.1.1c:** Analyze and describe how the different variables involved in homeostatic control mechanisms can be measured to determine homeostasis

### Classroom Exercise

#### Activity:

- Discuss the Concept of Homeostasis
- Describe the Negative feedback mechanism
- Describe the Positive Feedback mechanism
- List the Homeostatic Parameters and explain how they contribute to homeostasis?
  - Core temperature
  - Blood glucose
  - Iron levels
  - Copper regulation
  - Levels of blood gases
  - Blood oxygen content
  - Arterial blood pressure
  - Calcium levels
  - Sodium concentration
  - Potassium concentration
  - Fluid balance
  - Blood pH
  - Cerebrospinal fluid
  - Neurotransmission
  - Neuroendocrine system
  - Gene regulation
  - Energy balance

## Topic 2: Disease – Disruption of Homeostasis

### Performance Indicators

Students are expected to:

**12.1.1a:** Outline the clinical significance of what may go wrong in the body that disrupts maintenance of homeostasis resulting in fatal outcomes.

**12.1.1b:** Describe how diabetes as example of failed homeostatic control.

**12.1.1c:** Analyze and describe how the different variables involved in homeostatic control mechanisms can be measured to determine homeostasis

### Classroom Exercise

#### **Activity:**

- What is Disease?
- Describe diabetes and explain how it is A disease of Failed Homeostasis
  - Explain Normal Blood Sugar Regulation
  - Outline the Causes of Homeostasis to Evaluate Health
- How is Health Evaluated by Measuring Homeostasis?

## Topic 3: Systemic Response to Disease or Injury

### Performance Indicators

Students are expected to:

- 12.1.2a:** Examine how the human body responds to disease processes or injuries.
- 12.1.2b:** Evaluate the systemic inflammatory response and how inflammation is inhibited by neural pathways.
- 12.1.2c:** Analyze the basics of hormonal response to injury or disease states
- 12.1.2d:** Identify and describe the common inflammatory mediators and their roles
- 12.1.2b:** discover how cells respond to injury
- 12.1.2b:** analyze how cells mediate inflammatory response
- 12.1.2b:** Appreciate the immune and hematological system in responding to disease and injury state

### Classroom Exercise

#### Activity:

- Describe what Systemic Response to Injury is
- Explain the importance of The Systemic Inflammatory Response Syndrome (SIRS)
- Describe the Reflex Inhibition of Inflammation and explain the:
  - Afferent Signals to the Brain
  - Cholinergic Anti-Inflammatory Pathways
- Hormonal Response to Injury
  - Hormone Signaling Pathways
  - Adrenocorticotrophic Hormone
  - Cortisol and Glucocorticoids
  - Macrophage Inhibitory Factor
  - Growth Hormones and Insulin-like Growth Factors
  - Catecholamines
  - Aldosterone
  - Insulin
  - Acute Phase Proteins
- Mediators of Inflammation
  - Cytokines
  - Heat Shock Proteins
  - Reactive Oxygen Metabolites
  - Eicosanoid
  - Fatty Acid Metabolites
  - Kallikrein-Kinin System
  - Serotonin
  - Histamine
- Cellular Response to Injury
  - Gene Expression and Regulation
  - Cell Signaling Pathways
- Cell-Mediated Inflammatory Response
  - Platelets
  - Lymphocytes and T-Cell Immunity
  - Eosinophils

- Mast Cells
- Monocytes
- Neutrophils

## Topic 4: Surgery: Dissection of Animals

### Performance Indicators

Students are expected to:

**12.1.4a:** Dissect an animal and label the structure and function of it correctly.

**12.1.4b:** Distinguish the major systems and their functional relationships in correlation to human body

**12.1.4c:** Appreciate the similarities and differences in structure and function of the dissected rat and a human body

**12.1.4d:** Appreciate the use of surgical tools in dissection and hands on technique in stitching.

### Laboratory Practical: Dissection of a Rat

#### Activity:

- Introduction
- Classification of the Rat (*Rattus Norvegicus*)
- Dissection
- Grading
- Glossary of Terms
- Rate External Anatomy
  - Procedure
- The Muscular and Skeletal System of the Rat
  - Procedure: Skinning the Rat
  - Procedure: Exposing the bones of the leg
  - Structure of the Head and Neck
- Rat Dissection – The Thoracic Organs
- The Abdominal Organs
- Urogenital System
- Excretory Organs
- The Reproductive Organs of the Male Rat
- The Reproductive Organs of the Female Rat



# SAMPLE ASSESSMENT

## Sample Assessment Methods

- Group Assignment
- Presentation
- Projects
- Written or Practical Quiz
- Written or Practical Test
- Practical Assignment
- Case Studies

## Sample Assessment Tasks

| Module                                  | Content Standard                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Topic                                                                                                                                                                                            | Assessment Tasks            |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>12.1: Health and Medical Biology</b> | <p><b>12.1a:</b> Identify and describe the parameters and mechanisms that regulate homeostasis within the body.</p> <p><b>12.1b:</b> Evaluate and describe the disease processes within the human body that disrupts homeostasis.</p> <p><b>12.1c:</b> Identify and describe the pathway and level of health care in Papua New Guinea</p> <p><b>12.1d:</b> Appreciate and describe personalized medicine as the future trend of medical and health care model.</p> <p><b>12.1e:</b> Appreciate the underlying concepts and future technologies in health and medical care.</p> | <ol style="list-style-type: none"> <li>1. Homeostasis</li> <li>2. Disease – Disruption or Injury</li> <li>3. Systemic Response to Disease or Injury</li> <li>4. Dissection of Animals</li> </ol> | Project + Written Quiz/Test |

## Sample Assessment Marking Guide

| Topic         | Assessment Task | Assessment Criteria                                                      | Total Marks |
|---------------|-----------------|--------------------------------------------------------------------------|-------------|
| 3. Proteomics | Written Quiz    | What is Biophysics?                                                      | 2.5         |
|               |                 | How is proteomics and genomics related?                                  | 2.5         |
|               |                 | What is the difference between a gene and a genome?                      | 4           |
|               |                 | List at least three applications of biophysics and briefly explain them. | 6           |

|                                                                                                 |                                                                      |                                                                                |     |
|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------|-----|
|                                                                                                 |                                                                      | List the main ingredients used in Sanger sequencing.                           | 2.5 |
|                                                                                                 |                                                                      | Briefly describe the Edman Degradation process.                                | 2.5 |
| Subtotal                                                                                        |                                                                      |                                                                                | 20  |
| Note that this is similar to the Experiment given in GR 12 STEM Chemistry; THEMA: BIOTECHNOLOGY |                                                                      |                                                                                |     |
| Module 12.1 (All three Topics)                                                                  | Project/Experiment                                                   | Purify Protein Sequencing                                                      | 20  |
|                                                                                                 |                                                                      | Determine the amino acid sequence of a protein                                 | 20  |
|                                                                                                 |                                                                      | Check if the protein sequence exist on a database of protein sequences online. | 10  |
| Subtotal                                                                                        |                                                                      |                                                                                | 50  |
| Module 12.1 (All three Topics)                                                                  | Written Test<br>(Multiple Choices, Short Answers, Calculations, etc) | Correctly answers all long questions                                           | 10  |
|                                                                                                 |                                                                      | Correctly answers all multiple choice                                          | 15  |
|                                                                                                 |                                                                      | Correctly answers short answer questions                                       | 5   |
|                                                                                                 |                                                                      | Subtotal                                                                       |     |
| Module TOTAL                                                                                    |                                                                      |                                                                                | 100 |

# PLANNING, REPORTING & RECORDING (PERFORMANCE CRITERIA)

## Assessment Planning

| Assessment Domains                                         | Modules       | Total Marks | % Weight of Modules |
|------------------------------------------------------------|---------------|-------------|---------------------|
|                                                            | 6             |             |                     |
| Theory Knowledge & skills – <b>Test</b>                    | Test (50)     | /50         | <b>20</b>           |
| Application of knowledge – <b>Assignments and projects</b> | Assign.1 (30) | /30         | <b>30</b>           |
| End of Term <b>Exam</b>                                    | (25 marks)    | /25         | <b>50</b>           |
| <b>Total marks for term one</b>                            |               |             | <b>100 marks</b>    |

## Assessment Recording

| Percentage Range | Achievement Level               |
|------------------|---------------------------------|
| 85 -100          | Very High Achievement (VHA)     |
| 65-84            | High Achievement (HA)           |
| 50-64            | Satisfactory Achievement (SA)   |
| 25-49            | Low Achievement (LA)            |
| 0-24             | Below Minimum Achievement (BMA) |

| # | Name<br>s   |       |    |             |    | Assessment Task |    |         |               | Total<br>Score<br>(/300) | Percentage Weight (%) |                |               |           | Achie<br>vemen<br>t Level<br>[VHA,<br>HA,<br>SA,LA] |
|---|-------------|-------|----|-------------|----|-----------------|----|---------|---------------|--------------------------|-----------------------|----------------|---------------|-----------|-----------------------------------------------------|
|   |             | Tests |    | Assignments |    |                 |    | Project | Exam<br>(/50) |                          | T<br>(20%<br>)        | A + P<br>(30%) | Exam<br>(50%) | Ttl<br>CA |                                                     |
|   |             |       |    |             |    |                 |    |         |               |                          |                       |                |               |           |                                                     |
| 1 | Bill<br>Joe | 45    | 47 | 20          | 21 | 22              | 23 | 55      | 45            | 278                      | 18.4                  | 25.88          | 45            | 89.2<br>8 | VHA                                                 |
| 2 | Cara<br>Den | 20    | 25 | 15          | 16 | 17              | 18 | 25      | 30            | 166                      | 9                     | 18.22          | 30            | 57.2<br>2 | SA                                                  |
| 3 | Lea<br>May  | 9     | 15 | 5           | 6  | 7               | 8  | 30      | 25            | 105                      | 4.8                   | 9.10           | 25            | 38.9      | LA                                                  |

## Assessment Reporting

| <b>Thema / Biotechnology / Module 12.1: Protein Sequencing</b><br><b>Broad Content Standard for Assignments / project</b><br><b>12.1 To determine the amino acid sequence of a protein</b> |                                                            |                                                                        |                                                                                    |                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------|
| <b>Performance Criteria</b>                                                                                                                                                                | <b>Determine Protein Sequence (15 marks)</b>               | <b>Prepare Lab Report (5 marks)</b>                                    | <b>Answers all Relative Questions to protein sequencing (5 marks)</b>              | <b>25 marks</b> |
| <b>Very High Achievement</b>                                                                                                                                                               | Accurately determine protein sequence<br>(13-15)           | Accurately writes a lab report with correct sections<br>(5)            | Accurately answers all relative questions to protein sequencing<br>(5)             | <b>(21-25)</b>  |
| <b>High Achievement</b>                                                                                                                                                                    | Partially determine protein sequence<br>(10-12)            | Partially writes a lab report with correct sections<br>(4)             | Partially answers all relative questions to protein sequencing<br>(4)              | <b>(16-20)</b>  |
| <b>Satisfactory Achievement</b>                                                                                                                                                            | Fairly determine protein sequence<br>(7-9)                 | Fairly writes a lab report with correct sections<br>(3)                | Impartially answers all relative questions to protein sequencing<br>(3)            | <b>(11-15)</b>  |
| <b>Low Achievement</b>                                                                                                                                                                     | Poorly determine protein sequence<br>(4-6)                 | Poorly writes a lab report with correct sections<br>(2)                | Poorly answers all relative questions to protein sequencing<br>(2)                 | <b>(6-10)</b>   |
| <b>Below minimum Achievement</b>                                                                                                                                                           | Needs improvement in determining protein sequence<br>(0-3) | Needs improvement in writing a lab report with correct sections<br>(1) | Needs improvement in answering all relative questions to protein sequencing<br>(1) | <b>(0-5)</b>    |